

Integrating Individual and Social Contexts into Self-Reflection Technologies

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ABSTRACT

Technology for promoting reflection can help people better understand their emotions and thought patterns, eventually motivating them to take action to adopt healthy or productive behaviors. However, existing work has often viewed users individualistically, addressing people's behaviors and emotions rather than recognizing the external factors that shape them (e.g., economic status, culture). We envision that the individualistic approaches can be extended and reimaged in ways that can consider such broader contexts. We believe such a shift in the design of interventions will help individuals reflect in a more holistic manner, supporting collaborative reflection processes that involve more than one person. With these aims in mind, we will discuss the two questions in our workshop. First, what individual and social contexts should HCI researchers

consider while promoting reflection? Second, what role can various forms of technology (e.g., just-in-time adaptive interventions, peer-support platforms) play in supporting and augmenting reflective practices? Through our workshop, we hope to bring together a community of multidisciplinary researchers and practitioners who aim to design and develop reflection interventions that are situated within the fabric of users' individual and social contexts.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

reflection, mental health, behavior change, social determinant, psychology

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1 INTRODUCTION

“Personal informatics systems for supporting health largely grew out of a ‘self’-centric orientation: self-tracking, self-reflection, self-knowledge, self-experimentation, self-improvement. Health management, however, even when self-driven, is inherently social and depends on a person’s direct relationships and broader socio-cultural contexts, as an emerging line of research is coming to recognize, study, and support.” [28]

HCI and psychology researchers have established many potential benefits of practicing reflection, whether it be through prompting people with questions, activities, or information. The aim of these applications is often to help individuals better understand or revise their beliefs and intended actions by cognitive processes based on their past experiences [4, 11]. For example, an individual can use a mobile application to reflect on their daily step count data to find ways to be more physically active [22]. An online chatbot can initiate a conversation with a user to help them reflect on a past stressful situation, enabling them to reevaluate their thoughts and emotions and find alternative points of view [27]. Even designing simple targeted questions from psychology literature can motivate a person to practice reflection on their own and recognize cognitive distortions [30].

Existing self-reflection techniques in HCI and psychology literature have largely viewed users individualistically, addressing their unique behaviors and habits (e.g., mood, step count). While this approach has shown reasonable success in engendering behavior change and self-awareness, users often complain that interventions often end up being insensitive to external factors within and outside of their control [6, 10]. Many systems prompt users to reflect on whether they have met their goals, but failure to achieve a goal may be due to broader social barriers. For instance, the inability to meet a step count target may be due to concerns that a person has about the safety of their neighborhood. When these contexts are unacknowledged, users can be prone to blame themselves, leading to counterproductive results [8].

The contexts that can influence a person’s ability to achieve goals and reflect on their behaviors can be quite broad. They can refer to dynamic factors that can change within a span of minutes or hours (e.g., physical proximity with peers and colleagues, individual workload, relationship status), or they can refer to static factors (e.g., gender, race, economic status). Recent research has touched on the importance of these factors to a degree, showing that the needs and expectations of people from interventions vary significantly depending on their economic background, social status, religion, and family background [7, 31, 32]. However, HCI literature is yet to fully engage in designing reflection interventions that address the diversity of external social contexts.

Recognizing and addressing these contexts might better help individuals by providing them with holistic view of their problems, supporting collaborative reflection processes that involve more than one person. One prominent line of work that supports collaborative reflection has focused on developing peer-support platforms

to connect people who are experiencing similar problems, allowing them to coach each other and provide psychological support [30]. A different line of work has focused on identifying *social determinants of mental health* (i.e., the conditions in which people are born and grow up) as they design targeted interventions [2, 7]. A reevaluation of the current research trends in promoting reflection and addressing various social contexts is necessary to form clear actionable plans for future work in this area. Our workshop aims to build a multidisciplinary community consisting of researchers to pioneer this challenge. We are motivated by the following two broad themes:

- What individual and social contexts should HCI researchers consider while promoting reflection?
- What role can various forms of technology (e.g., just-in-time adaptive interventions, peer-support platforms) play in supporting and augmenting reflective practices?

2 THEMES

Below, we discuss each of our themes by briefly describing relevant past works and outlining our goals.

2.1 Role of Individual and Social Contexts in Promoting Reflection

The importance of several individual and social contexts is already established in prior research on health and wellbeing [12, 13, 28]. For example, there is a body of literature that connects health and wellbeing with circadian rhythms [15, 26, 43] — the physical, mental, and behavioral patterns that follow a 24-hour cycle [42]. Poor mental health conditions may involve distortions of the typical rhythms. As an example, individuals may find themselves waking up, going to bed, eating, and socializing at inconsistent times from day-to-day. These disruptions can lead to a sense of being ungrounded and symptoms of depression. Health and wellbeing has also been linked to physical activity, movement, and social proximity [5]. For instance, lower levels of depression have been observed when individuals are more physically active [21, 41], spend more time outside of the house [37], and engage socially with others [39].

There is a separate line of work that examined social determinants in discussions of reflection and mental health [2, 7]. Social determinants refer to the conditions in which people are “born, grow, live, work, and age” [25]. Inequities in social determinants can lead to social gradients [16], which describe situations when people with higher social status have longer life expectancy and fewer health risks compared to those with lower social status.

A number of studies have linked poor mental health with financial hardships [9, 23, 35]. For example, Katz-Wise et al. [17] found strong association between lower income and symptoms of depression and suicidal ideation among transgender individuals living in the USA. Starkey et al. [40] also observed similar results for African women and suggested that treatments strategies should be adapted based on one’s level of financial distress. A British study by Pevalin et al. [33] found that consistently living in poor housing conditions can worsen mental health of youth and adults. A relevant body of work [31, 32, 36] also explored the role of faith and spirituality in promoting healthcare interventions; for example, many black

Christians expressed that when they are stressed, they look for spiritual support in the form of prayers or Bible study sessions [32].

We are interested in exploring how these diverse individual and social factors impact one's ability to engage in reflection and how reflection interventions should adapt according to them. Moreover, we aim to explore the cross-cultural differences in needs and expectations from such interventions, particularly because there are not many relevant research works in the context of the Global South. By facilitating conversations among people from various disciplines and backgrounds through our workshop, we aim to gain insights into these issues.

2.2 Promise of Technology in Supporting Reflection

Numerous software and apps have been developed in the last two decades to promote reflection in diverse settings, ranging from self-monitoring tools to just-in-time adaptive interventions. Self-monitoring tools have enabled people to reflect on their beliefs and behaviors at their own convenience. These tools usually attempt to increase emotional self-awareness (ESA) — an individual's ability to recognize their own emotions [18]. For example, Bakker and Rickard [3] developed MoodPrism, a mobile application that prompts participants to answer a few daily questions about their mood so that people can recognize trends in their mood history. A similar platform by Bhattacharjee et al. [7] enabled participants to review their conversation history through graphs and charts. A study by Kauer et al. [19] showed that monitoring mood, stress, and coping strategies can reduce one's ESA and reduce symptoms of depression.

Some recent works have attempted to support collaborative reflection processes by virtually connecting people from diverse backgrounds [34]. The guided chat tool by O'Leary et al. [30] supports collaborative reflection by providing chat partners a sequence of reflective prompts based on behavioral chain analysis [38]. Other works have tried using shared whiteboards [20] and pictures as memory aids [14]. More recently, social media channels have also supported collaborative reflection, where people interested in similar topics can engage in conversation with each other and collectively learn from each other's experience [4].

In parallel to these efforts, researchers have designed various forms of technology to detect and adapt to a person's contextual factors. A notable line of work involves the development of just-in-time adaptive interventions (JITAI) that use real-time information about users to deliver the *"the right type/amount of support, at the right time, by adapting to an individual's changing internal and contextual state"* [29]. There are many ways researchers can collect in-the-moment information about users' mood and activity in scalable ways. For example, the accelerometer built into people's smartphones can be used as a proxy to monitor activities and physical movement [24]. Moreover, location sensors like the GPS and WiFi can also reveal significant information about where people spend their time [1].

Through our workshop, we aim to collectively identify promising directions for novel platforms that can support reflection by addressing diverse individual and social factors. These platforms may connect individuals via social media groups and peer support

platforms to make reflective practices more collaborative, or they may automatically identify people's context using mechanisms like sensors or momentary forms. We also notice that although literature has established the role of factors like racism, gender discrimination, or inequality in shaping one's health and wellbeing, HCI researchers often find it challenging to integrate these factors with technology. We believe our workshop will allow researchers to discuss these issues and form clear, actionable plans for the future.

3 ORGANIZERS

The organizers have conducted research in human-computer interaction, machine learning, psychology, social science, medical science, and communication. They are from different social and cultural backgrounds and have experiences of interacting and working with people from diverse backgrounds. We have collective experiences of organizing CHI workshops. Below we provide brief information about each organizer.

Ananya Bhattacharjee (primary contact) is a Ph.D. student in the Department of Computer Science at the University of Toronto. His research focuses on building technology to help people manage mental health. He conducts research in different social and cultural backgrounds: he has collaborated with Mental Health America to understand users' needs and design interventions accordingly; he conducted field studies in Bangladesh to understand the alignment between existing mental health technologies and local people's customs and values. His research works have been published in top venues like CHI, CSCW, TOCHI, and JMIR.

Dana Kulzhabayeva is a Ph.D. student in the Department of Psychology at the University of Toronto. She is interested in judgment and decision making. She is leading a project to design technology-based nudges to facilitate positive health behaviors; she has collaborated with HCI researchers to understand people's behaviors when interacting with digital mental health tools.

Mohi Reza is a Ph.D. student in the Department of Computer Science at the University of Toronto. His research is at the intersection of HCI and education technology. He designs tools to facilitate learning science research in online courses using adaptive field experiments, and investigates the effects of various interventions intended to help students in multiple areas of their lives (e.g., planning, growth mindset, stress management).

Harsh Kumar is a Ph.D. student in the Department of Computer Science at the University of Toronto. His research interests lie in Human Computer Interaction, Machine Learning, Natural Language Processing and their intersection with Social Sciences. He conducts research in designing text messaging systems to prompt reflection and manage mental health. He also explores the potential of developing mental health chatbots.

Eunchae Seong is an undergraduate research assistant in the Department of Computer Science at the University of Toronto. Her main interests in research are human-computer interaction for education and psychology. She has been participating in mental health and education-related projects as a data analyst and programmer.

Xuening Wu is an undergraduate research assistant in the Department of Psychology at the University of Toronto. She is interested in social psychology and its applications. She has participated in research projects related to social psychology and human-computer interaction. She keeps seeking ways to apply social psychology to design more personalized digital mental health tools.

Mohammad Rashidujjaman Rifat is a Ph.D. candidate in the Department of Computer Science at the University of Toronto. His research is at the intersection of HCI, ICTD, and faith. He conducts qualitative studies and computational analysis to explore faith-based values, rationality, and politics; and designs technologies to mitigate faith-based intolerance and make technologies more faith inclusive.

Robert Bowman is a Ph.D. candidate in School of Computer Science and Statistics at Trinity College Dublin. His research concerns the design of conversational user interfaces to support people with their mental health during their everyday lives.

Rachel Kornfield is a Research Assistant Professor in the Department of Preventive Medicine and the Center for Behavioral Intervention Technologies in Northwestern University's Feinberg School of Medicine. She earned her Ph.D. from the University of Wisconsin-Madison. Her program of research focuses on the role of supportive communication within digital mental health interventions, including communication with peers, mental health professionals, and automated messaging systems.

Alex Mariakakis is an Assistant Professor in the Department of Computer Science at the University of Toronto and an Affiliate Scientist at Techna. He completed his Ph.D. at the University of Washington. He runs the Computational Health and Interaction (CHAI) lab, which leverages ubiquitous and emergent technologies to address problems related to people's health and wellbeing.

Syed Ishtiaque Ahmed is an Assistant Professor of Computer Science at University of Toronto. He conducts research in the intersection between Human-Computer Interaction (HCI) and Information and Communication Technology and Development (ICTD). He received his Ph.D. in Information Science from Cornell University in 2017. He established the first HCI research lab in Bangladesh in 2009, and still maintains it. His research work is built around the concept of 'voice' that connects various branches of political philosophy to technology intervention.

Munmun De Choudhury is an Associate Professor in the School of Interactive Computing at Georgia Tech. She received her Ph.D. from the Department of Computer Science at Arizona State University, Tempe. She founded and directs the Social Dynamics and Wellbeing Lab (SocWeB Lab). Her research interests lie in the intersection of computer science and social science. Her research has pioneered the computational use of social media data for mental health.

Gavin Doherty is an Associate Professor in the School of Computer Science and Statistics, at Trinity College, Dublin, where he leads the Health Technology Design Group. He received his D.Phil at the University of York. Much of his research concerns the design of mental health interventions, with a particular focus on engagement.

Mary Czerwinski is a Partner Research Manager of the Human Understanding and Empathy group at the Microsoft Research.

Her research focuses primarily on information worker task management, health and wellness for individuals and groups. She has received many awards for her exceptional work in applying psychological principles to the design and understanding of human-computer interaction.

Joseph Jay Williams is an Assistant Professor in the Department of Computer Science at the University of Toronto, cross-appointed by the Department of Psychology and Statistics. His research focuses on combining Bayesian statistics and Machine Learning algorithms to conduct randomized experiments in mental health and educational settings. He leads and founded the Intelligent Adaptive Interventions research group. His work has been published in top venues and received great media exposure.

4 WEBSITE

The link to the website is

<https://sites.google.com/view/reflectionchiworkshop2023>. We will update the website with the call for participation, information about attendance, and the workshop agenda by December 15, 2022.

5 PRE-WORKSHOP PLANS

We will invite participants from various disciplines, including human-computer interaction, clinical psychology, experimental psychology, social science, and medical science. Our participants will come from both the Global North and South. Based on the composition of our organizing committee alone, we expect to have contributors and attendees from the USA, Canada, the UK, Trinidad, India, Pakistan, and Bangladesh. We will distribute the call for participation via relevant professional mailing lists, social media outlets, and other research community channels to publicize the workshop beyond our networks and reach the widest audience possible. We hope to recruit 25–30 participants for the workshop.

We will ask participants to submit position papers (2–6 pages long in the ACM Master Article Template) relevant to the workshop themes. The organizing team will review all submissions. The review criteria will be based on the proposals' fit with the workshop goals and contribution to thematic discussions. All accepted papers and presentations will be made available on the workshop website. To enhance asynchronous engagement, we will invite participants to a Slack workspace where they can engage in conversations and connect with each other.

6 PRE-WORKSHOP DOCUMENTS

To encourage participants to learn more about each other's work before the conference, we will ask everyone to send us a 1-minute video clip and a short paragraph of text to introduce themselves. We will require the videos to have closed captions to ensure they are accessible to those who require them. We will upload these materials to an #introductions channel in our Slack workspace and invite participants to engage with each other before the workshop. By doing this, participants will be able to get to know each other and begin thinking about the kinds of discussions they can have during the workshop.

Table 1: The workshop structure

Time	Activity
9:00 - 9:10	Welcome and Introduction
9:10 - 10:00	Opening Keynote by <Name>. Topic:
10:00 - 10:30	Coffee Break and Networking Session
10:30 - 11:30	Paper Presentation: Theme 1
11:30 - 12:30	Paper Presentation: Theme 2
12:30 - 13:30	Lunch Break
13:30 - 14:30	Problem Pitch Mentoring Session
14:30 - 15:30	Panel Discussion
15:30 - 16:20	Closing Keynote by <Name>. Topic:
16:20 - 16:30	Closing Remarks

7 THE WORKSHOP DAY

We will conduct the workshop in a hybrid format; some organizers and participants will be in-person, and others will join remotely. Table 1 contains information on the tentative schedule for the day-long workshop. We will begin with introductory remarks from the organizers, followed by a keynote presentation. After the opening keynote session, there will be a networking session and coffee break. Then, we will have two sessions for paper presentations. Following the presentations and a lunch break, there will be a mentorship session where early-stage researchers will have an opportunity to connect with senior researchers and receive guidance on topics relevant to the workshop. After a panel discussion with the organizers, we will conclude the workshop with a keynote and closing remarks.

8 POST-WORKSHOP PLANS

To have a public record of the workshop, the position papers will remain available on the workshop website. We shall also publish a summary of the discussions that occur during the workshop itself. To further consolidate and disseminate the ideas from those discussions we will write a paper that proposes a research agenda for context-sensitive digital reflection technologies. This research agenda will form the basis of a white paper that we shall organize to encourage further research on the ideas discussed during the workshop.

9 ACCESSIBILITY

To make our workshop accessible, we will request participants to share with us any accessibility requirements they may have in advance and put in place any necessary accommodations. As mentioned earlier, introductory videos will be required to have closed captions. We will enable auto-generated captions in our video-conferencing tool for remote attendees, and include alternative text on our workshop website that allows screen readers to describe figures and other graphics.

10 CALL FOR PARTICIPATION

Technology for promoting reflection can help people understand their emotions and thought patterns, motivating them to take action to adopt healthy behaviors. The uptake and efficacy of these technologies are impacted by many individual and social factors —

economic status, culture, faith, etc. — yet prior work has only explored a handful of these influences. During our interactive hybrid workshop, we will examine the role of these diverse contextual factors, how they impact user engagement with digital interventions, and how technology can better address them. We invite multidisciplinary researchers and practitioners to submit position papers under the following themes:

- **Theme 1:** What individual and social contexts should HCI researchers consider while promoting reflection?
- **Theme 2:** What role can various forms of technology (e.g., just-in-time adaptive interventions, peer-support platforms) play in supporting and augmenting reflective practices?

Submission Details:

- **Submission deadline:** February 9, 2023
- **Notification of acceptance:** February 23, 2023
- **Page limit:** 2-6 pages (including references)
- **Template:** ACM Master Article Submission Templates, single column (<https://chi2023.acm.org/for-authors/presenting/papers/chi-publication-formats/>)
- **Selection criteria:** Contribution to workshop’s themes, quality of presentation, and potential to stimulate discussions.
- **Submission:** Email to reflectionchiworkshop@gmail.com with the subject line “CHI 2023 Workshop”. Upon acceptance, at least one author must attend the workshop, prepare a three-minute-long video presentation, and register for the workshop at least one day before the conference. Accepted papers will be archived on the workshop’s website.
- **Website:** <https://sites.google.com/view/reflectionchiworkshop2023>.

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REFERENCES

- [1] Saeed Abdullah, Mark Matthews, Ellen Frank, Gavin Doherty, Geri Gay, and Tanzeem Choudhury. 2016. Automatic detection of social rhythms in bipolar disorder. *Journal of the American Medical Informatics Association* 23, 3 (2016), 538–543.
- [2] Jessica Allen, Reuben Balfour, Ruth Bell, and Michael Marmot. 2014. Social determinants of mental health. *International review of psychiatry* 26, 4 (2014), 392–407.
- [3] David Bakker and Nikki Rickard. 2018. Engagement in mobile phone app for self-monitoring of emotional wellbeing predicts changes in mental health: MoodPrism. *Journal of affective disorders* 227 (2018), 432–442.
- [4] Amit Baumel, Amanda Tinkelman, Nandita Mathur, and John M Kane. 2018. Digital peer-support platform (7Cups) as an adjunct treatment for women with postpartum depression: feasibility, acceptability, and preliminary efficacy study. *JMIR mHealth and uHealth* 6, 2 (2018), e9482.
- [5] Alexandra H Bettis, Taylor A Burke, Jacqueline Nesi, and Richard T Liu. 2022. Digital technologies for emotion-regulation assessment and intervention: A conceptual review. *Clinical Psychological Science* 10, 1 (2022), 3–26.
- [6] Ananya Bhattacharjee, Mohammad Ruhul Amin, Yeshim Iqbal, and Syed Ishtiaque Ahmed. 2022. Connecting Mental Health with Sustainable Development Goals: Insights from Call Data of a Telephone Crisis Helpline in Bangladesh. (2022).

- [7] Ananya Bhattacharjee, SM Taiabul Haque, Md Abdul Hady, SM Raihanul Alam, Mashfiqui Rabbi, Muhammad Ashad Kabir, Syed Ishtiaque Ahmed, et al. 2021. Understanding the Social Determinants of Mental Health of Undergraduate Students in Bangladesh: Interview Study. *JMIR Formative Research* 5, 11 (2021), e27114.
- [8] Ananya Bhattacharjee, Jiayu Pang, Angelina Liu, Alex Mariakakis, and Joseph Jay Williams. 2022. Design Implications for One-Way Text Messaging Services that Support Psychological Wellbeing. (In Press) *ACM Transactions on Computer-Human Interaction (TOCHI)* (2022).
- [9] Anna Brydsten, Anne Hammarström, and Miguel San Sebastian. 2018. Health inequalities between employed and unemployed in northern Sweden: a decomposition analysis of social determinants for mental health. *International Journal for Equity in Health* 17, 1 (2018), 1–11.
- [10] Felipe González Castro, Manuel Barrera, and Charles R Martinez. 2004. The cultural adaptation of prevention interventions: Resolving tensions between fidelity and fit. *Prevention science* 5, 1 (2004), 41–45.
- [11] J Dewey. 1933. How We Think. A Restatement of the Relation of Reflective Thinking to the Educative Process, Boston etc.(DC Heath and Company) 1933. (1933).
- [12] Anind K Dey. 2001. Understanding and using context. *Personal and ubiquitous computing* 5, 1 (2001), 4–7.
- [13] Paul Dourish. 2004. What we talk about when we talk about context. *Personal and ubiquitous computing* 8, 1 (2004), 19–30.
- [14] Rowanne Fleck and Geraldine Fitzpatrick. 2010. Reflecting on reflection: framing a design landscape. In *Proceedings of the 22nd Conference of the Computer-Human Interaction Special Interest Group of Australia on Computer-Human Interaction*. 216–223.
- [15] Russell G Foster, Stuart N Peirson, Katharina Wulff, Eva Winnebeck, Céline Vetter, and Till Roenneberg. 2013. Sleep and circadian rhythm disruption in social jetlag and mental illness. *Progress in molecular biology and translational science* 119 (2013), 325–346.
- [16] Olivia Souza Honório, Milene Cristine Pessoa, Lucia Helena Almeida Gratão, Luana Lara Rocha, Inês Rugani Ribeiro de Castro, Daniela Silva Canella, Paula Martins Horta, and Larissa Loures Mendes. 2021. Social inequalities in the surrounding areas of food deserts and food swamps in a Brazilian metropolis. *International journal for equity in health* 20, 1 (2021), 1–8.
- [17] Sabra L Katz-Wise, Sari L Reisner, Jaclyn M White Hughto, and Stephanie L Budge. 2017. Self-reported changes in attractions and social determinants of mental health in transgender adults. *Archives of sexual behavior* 46, 5 (2017), 1425–1439.
- [18] SD Kauer. 2012. Emotional Self-Awareness Scale. *Journal of Medical Internet Research* 14, 3 (2012), e67.
- [19] Sylvia Deidre Kauer, Sophie Caroline Reid, Alexander Hew Dale Crooke, Angela Khor, Stephen John Charles Hearps, Anthony Francis Jorm, Lena Sanci, and George Patton. 2012. Self-monitoring using mobile phones in the early stages of adolescent depression: randomized controlled trial. *Journal of medical Internet research* 14, 3 (2012), e1858.
- [20] Dongsik Kim and Seunghee Lee. 2002. Designing collaborative reflection supporting tools in e-project-based learning environments. *Journal of Interactive Learning Research* 13, 4 (2002), 375–392.
- [21] Seoyoun Kim, Hyunwoo Yoon, Patricia Morton, and Yuri Jang. 2022. Longitudinal links between behavioral activation coping strategies and depressive symptoms of US adults living alone during the COVID-19 pandemic. *Plos one* 17, 5 (2022), e0267948.
- [22] Rafal Kocielnik, Lillian Xiao, Daniel Avrahami, and Gary Hsieh. 2018. Reflection companion: a conversational system for engaging users in reflection on physical activity. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 2 (2018), 1–26.
- [23] Susanne Sundell Lecerof, Martin Stafström, Ragnar Westerling, and Per-Olof Östergren. 2015. Does social capital protect mental health among migrants in Sweden? *Health promotion international* 31, 3 (2015), 644–652.
- [24] Yuanchao Ma, Bin Xu, Yin Bai, Guodong Sun, and Run Zhu. 2012. Daily mood assessment based on mobile phone sensing. In *2012 ninth international conference on wearable and implantable body sensor networks*. IEEE, 142–147.
- [25] Michael Marmot. 2011. Global action on social determinants of health. , 702–702 pages.
- [26] Mark Matthews, Erin Carroll, Saeed Abdullah, Jaime Snyder, Matthew Kay, Tanzeem Choudhury, Geri Gay, and Julie Kientz. 2014. Biological rhythms and technology. In *CHI'14 Extended Abstracts on Human Factors in Computing Systems*. 123–126.
- [27] Robert R Morris, Stephen M Schueller, and Rosalind W Picard. 2015. Efficacy of a web-based, crowdsourced peer-to-peer cognitive reappraisal platform for depression: randomized controlled trial. *Journal of medical Internet research* 17, 3 (2015), e4167.
- [28] Elizabeth L. Murnane, Tara G. Walker, Beck Tench, Stephen Volda, and Jaime Snyder. 2018. Personal Informatics in Interpersonal Contexts: Towards the Design of Technology That Supports the Social Ecologies of Long-Term Mental Health Management. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW, Article 127 (nov 2018), 27 pages. <https://doi.org/10.1145/3274396>
- [29] Inbal Nahum-Shani, Shawna N Smith, Bonnie J Spring, Linda M Collins, Katie Witkiewitz, Ambuj Tewari, and Susan A Murphy. 2018. Just-in-time adaptive interventions (JITAI) in mobile health: key components and design principles for ongoing health behavior support. *Annals of Behavioral Medicine* 52, 6 (2018), 446–462.
- [30] Kathleen O'Leary, Stephen M Schueller, Jacob O Wobbrock, and Wanda Pratt. 2018. "Suddenly, we got to become therapists for each other" Designing Peer Support Chats for Mental Health. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [31] Teresa K. O'Leary, Elizabeth Stowell, Jessica A. Hoffman, Michael Paasche-Orlow, Timothy Bickmore, and Andrea G. Parker. 2021. Examining the Intersections of Race, Religion & Community Technologies: A Photovoice Study. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 698, 19 pages. <https://doi.org/10.1145/3411764.3445418>
- [32] Teresa K. O'Leary, Elizabeth Stowell, Darley Sackitey, Hye Sun Yun, David Wright, Michael Paasche-Orlow, Timothy Bickmore, and Andrea G. Parker. 2022. Church after Sunday: Supporting Everyday Well-Being through Techno-Spiritual Health Interventions. *Interactions* 29, 4 (jun 2022), 90–93. <https://doi.org/10.1145/3542838>
- [33] David J Pevalin, Aaron Reeves, Emma Baker, and Rebecca Bentley. 2017. The impact of persistent poor housing conditions on mental health: A longitudinal population-based study. *Preventive medicine* 105 (2017), 304–310.
- [34] Michael Prilla, Oliver Blunk, and Irene-Angelica Chounta. 2020. How does collaborative reflection unfold in online communities? An analysis of two data sets. *Computer Supported Cooperative Work (CSCW)* 29, 6 (2020), 697–741.
- [35] Nadine Reibling, Jason Beckfield, Tim Huijts, Alexander Schmidt-Catran, Katie H Thomson, and Claus Wendt. 2017. Depressed during the depression: has the economic crisis affected mental health inequalities in Europe? Findings from the European Social Survey (2014) special module on the determinants of health. *The European Journal of Public Health* 27, suppl_1 (2017), 47–54.
- [36] Mohammad Rashidujjaman Rifat, Firaz Ahmed Peer, Hawra Rabaan, Nusrat Jahan Mim, Maryam Mustafa, Kentaro Toyama, Robert B Markum, Elizabeth Buie, Jessica Hammer, Sharifa Sultana, et al. 2022. Integrating Religion, Faith, and Spirituality in HCI. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–6.
- [37] Darius A Rohani, Maria Faurholt-Jepsen, Lars Vedel Kessing, and Jakob E Bardram. 2018. Correlations between objective behavioral features collected from mobile and wearable devices and depressive mood symptoms in patients with affective disorders: systematic review. *JMIR mHealth and uHealth* 6, 8 (2018), e9691.
- [38] Michael B Schiffer. 1975. Behavioral chain analysis: Activities, organization, and the use of space. *Fieldiana. Anthropology* 65 (1975), 103–119.
- [39] Nili Solomonov, Jennifer N Bress, Jo Anne Sirey, Faith M Gunning, Christoph Flückiger, Patrick J Raue, Patricia A Areán, and George S Alexopoulos. 2019. Engagement in socially and interpersonally rewarding activities as a predictor of outcome in "Engage" behavioral activation therapy for late-life depression. *The American Journal of Geriatric Psychiatry* 27, 6 (2019), 571–578.
- [40] Angelica JoNel Starkey, Christopher R Keane, Martha Ann Terry, John H Marx, and Edmund M Ricci. 2013. Financial distress and depressive symptoms among African American women: identifying financial priorities and needs and why it matters for mental health. *Journal of Urban Health* 90, 1 (2013), 83–100.
- [41] Kristin L Szuhany and Michael W Otto. 2020. Efficacy evaluation of exercise as an augmentation strategy to brief behavioral activation treatment for depression: a randomized pilot trial. *Cognitive behaviour therapy* 49, 3 (2020), 228–241.
- [42] Martha Hotz Vitaterna, Joseph S Takahashi, and Fred W Turek. 2001. Overview of circadian rhythms. *Alcohol research & health* 25, 2 (2001), 85.
- [43] William H Walker, James C Walton, A Courtney DeVries, and Randy J Nelson. 2020. Circadian rhythm disruption and mental health. *Translational psychiatry* 10, 1 (2020), 1–13.